

Submitting for  and just to see what can happen.

Problem 1.1 / Example 1.7

For every k rows, there are $4k$ cards that show up. By Pigeonhole, there are at least k distinct ranks. Therefore, Hall's criterion is met. By Hall's Theorem we are done.

Problem 1.2 / Example 1.8

Every set of k regions on one unit square has area $\frac{k}{n}$. By Pigeonhole, this must match to at least k regions on the other square. By Hall's we are done.

Problem 1.3 / Example 1.9

This is possible.

We show it is possible match each possible choice of four candies with a choice of 4 candies distinct from the four rare ones that Eric will report such that every report by Eric is different.

We may think of a graph with $\binom{12}{4}$ vertices on the left side representing all choices of 4 rare coins from 12 and $\binom{12}{4}$ vertices on the right side representing all choices of 4 coins Eric will report. For each choice of 4 coins on the left side, we draw an edge from it to each of the $\binom{8}{4}$ choices of 4 different coins on the right side.

Clearly, this graph is regular since by symmetry, the degree of each vertex on the right side is also $\binom{8}{4}$. Therefore, we are done.